

AWS EC2 Deployment Guide

YouTube Quiz Generator

Step-by-Step Guide for Free Tier Deployment

This guide walks you through deploying the YouTube Quiz Generator on an AWS EC2 t2.micro instance (12 months free).

Why AWS instead of Render?

Render's free tier drops connections after 30 seconds of inactivity, causing 'connection was lost' errors during quiz generation. On EC2, you control the timeout settings (300s), eliminating this issue.

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1. Prerequisites

Before you begin, you need the following:

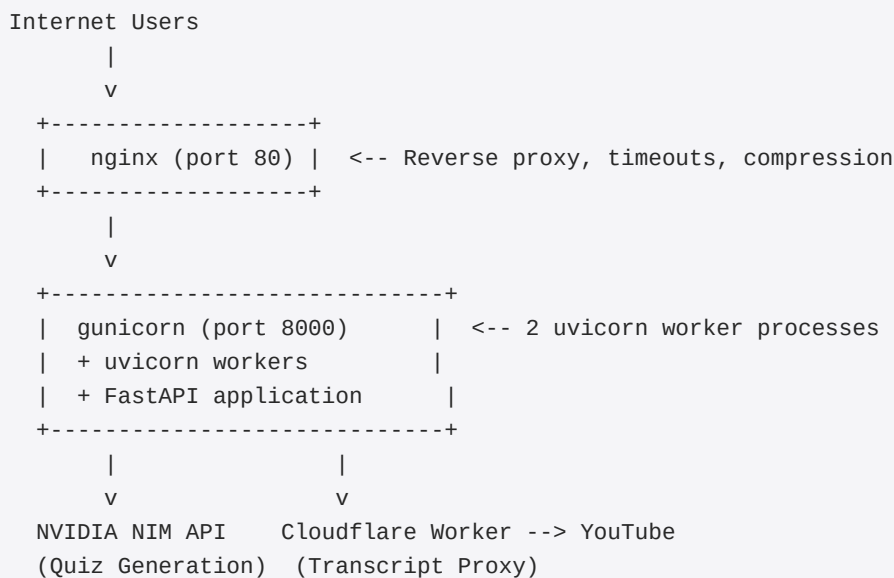
- An AWS account (free tier eligible) - sign up at <https://aws.amazon.com/>
- A NVIDIA NIM API key - get one free at <https://build.nvidia.com/>
- An SSH client (Terminal on Mac/Linux, PuTTY or WSL on Windows)
- A web browser to access the AWS EC2 console

Cost

AWS EC2 t2.micro is FREE for 12 months (750 hours/month). After 12 months, it costs ~\$8.50/month. Data transfer is also free within the first 100GB/month.

2. Architecture Overview

The deployment architecture on AWS EC2:



Problem	Render Free Tier	AWS EC2 (This Setup)
Connection timeout	30 seconds (fixed)	300 seconds (configurable)
Proxy buffering	Enabled (kills SSE)	Disabled for SSE endpoint
Keepalive control	None	Full control (12s pings)
RAM	512 MB	1 GB + 2 GB swap
Cold starts	30-60 seconds	None (always-on)
Cost	Free forever	Free 12 months, ~\$8.50/mo after

3. Launch an EC2 Instance

Follow these steps to create your EC2 instance on AWS:

Step 1: Log in to AWS Console

Go to <https://console.aws.amazon.com/ec2/> and sign in with your AWS account.

Step 2: Launch a New Instance

Click the 'Launch Instance' button (orange button, top right).

Step 3: Configure Instance Settings

Name and Tags

Enter a name like 'youtube-quiz-generator'.

AMI (Operating System)

Select 'Ubuntu 22.04 LTS' or 'Ubuntu 24.04 LTS' from the Quick Start list. This should be the default option.

Instance Type

Select 't2.micro' (Free tier eligible). If you see 't3.micro' as the default, change it to t2.micro - both work, but t2.micro is guaranteed free tier.

Step 4: Create a Key Pair (for SSH access)

Under 'Key pair (login)', click 'Create new key pair'.

- Name: quiz-generator-key (or any name you prefer)
- Type: RSA
- Format: .pem (for Mac/Linux/WSL) or .ppk (for PuTTY on Windows)

Click 'Create key pair' and save the downloaded file. Move it to a secure location (~/.ssh/ on Mac/Linux).

```
# Mac/Linux: set correct permissions
chmod 400 ~/Downloads/quiz-generator-key.pem
mv ~/Downloads/quiz-generator-key.pem ~/.ssh/
```

Step 5: Configure Security Group (Network Settings)

Under 'Network settings', configure the firewall:

Type	Protocol	Port	Source	Why
SSH	TCP	22	Your IP	SSH access to manage the server
HTTP	TCP	80	0.0.0.0/0	Web app access from browsers
HTTPS	TCP	443	0.0.0.0/0	HTTPS (for certbot later)

Security Tip

For SSH, restricting the source to 'Your IP' is more secure than 0.0.0.0/0. If your IP changes, you can update the security group rule later.

Step 6: Add User Data (for automated setup)

Scroll down to 'Advanced details' and find the 'User data' text box at the bottom. This is where you paste the automated setup script.

Option A vs B vs C

Option A (Recommended): Paste userdata.sh into the User data field - fully automated.

Option B: Skip user data, SSH in after launch, and run `deploy.sh` manually.

Option C: Follow the step-by-step manual setup in Section 6.

Step 7: Launch!

Click 'Launch instance'. Wait 3-5 minutes for the instance to start and the setup script to complete. Then proceed to Section 7 to set your API key.

4. Option A: Automated Deployment (Recommended)

This is the fastest way to deploy. The entire server is configured automatically when the EC2 instance first boots.

Step 1: Get the User Data Script

The file is located at `deploy/aws/userdata.sh` in the repository. You can either:

- Open it from the cloned repo
- Or [view it on GitHub](https://github.com/Nikhilramswaminaidu/youtube_video_quiz_generator/blob/main/deploy/aws/userdata.sh):
https://github.com/Nikhilramswaminaidu/youtube_video_quiz_generator/blob/main/deploy/aws/userdata.sh

Copy the ENTIRE contents of `userdata.sh` to your clipboard.

Step 2: Paste into EC2 User Data

During EC2 instance launch (Step 6 above), paste the entire `userdata.sh` script into the 'User data' text box under 'Advanced details'.

```
# The user data field looks like this:
#
# +-----+
# | User data |
# |          |
# | #!/bin/bash |
# | # Paste the userdata.sh content here |
# | set -euo pipefail |
# | ... |
# +-----+
```

Step 3: Launch and Wait

Launch the instance. The setup script runs automatically on first boot. It takes approximately 3-5 minutes to complete. The script will:

- Update system packages and install Python 3.11, nginx, ffmpeg, git
- Create a 2GB swap file (t2.micro has only 1GB RAM)
- Create a 'quizapp' system user
- Clone the repository into `/opt/youtube-quiz-generator`
- Set up a Python virtual environment and install all dependencies
- Configure nginx as a reverse proxy with SSE-optimized timeouts
- Install a systemd service for auto-restart on crash
- Start both the app and nginx

Step 4: Set Your API Key (REQUIRED)

The app won't work without your NVIDIA API key. SSH into your instance and edit the `.env` file:

```
# SSH into your instance (replace with your key and IP)
ssh -i ~/.ssh/quiz-generator-key.pem ubuntu@<YOUR_EC2_PUBLIC_IP>

# Edit the .env file
sudo nano /opt/youtube-quiz-generator/.env

# Replace the placeholder with your actual key:
#   NVIDIA_API_KEY=nvapi-paste-your-key-here
# becomes:
#   NVIDIA_API_KEY=nvapi-your-actual-key-here

# Save and exit (Ctrl+X, Y, Enter)

# Restart the app to pick up the new key
sudo systemctl restart quiz-generator
```

Step 5: Visit Your App

Open your browser and go to:

```
http://<YOUR_EC2_PUBLIC_IP>/
```

You can find your EC2 public IP in the AWS Console under your instance details, or by running this command on the instance:

```
curl http://169.254.169.254/latest/meta-data/public-ipv4
```

5. Option B: Manual Deployment Script

If you already have a running EC2 instance without user data, use the deploy.sh script.

Step 1: SSH Into Your Instance

```
ssh -i ~/.ssh/quiz-generator-key.pem ubuntu@<YOUR_EC2_PUBLIC_IP>
```

Step 2: Clone the Repository

```
git clone https://github.com/Nikhilramswaminaidu/youtube_video_quiz_generator.git
cd youtube_video_quiz_generator
```

Step 3: Run the Deploy Script

```
sudo bash deploy/aws/deploy.sh
```

The script runs through the same steps as userdata.sh but with colored output and progress indicators. It takes 3-5 minutes to complete.

Step 4: Set Your API Key

```
sudo nano /opt/youtube-quiz-generator/.env
# Replace NVIDIA_API_KEY=nvapi-paste-your-key-here
# with your actual key
sudo systemctl restart quiz-generator
```

6. Option C: Step-by-Step Manual Setup

If you want full control over every step, follow this guide.

Step 1: SSH Into Your Instance

```
ssh -i ~/.ssh/quiz-generator-key.pem ubuntu@<YOUR_EC2_PUBLIC_IP>
```

Step 2: Update System & Install Packages

```
sudo apt-get update && sudo apt-get upgrade -y
sudo apt-get install -y curl wget git software-properties-common \
    build-essential libssl-dev zlib1g-dev nginx ffmpeg

# Install Python 3.11
sudo add-apt-repository -y ppa:deadsnakes/ppa
sudo apt-get update
sudo apt-get install -y python3.11 python3.11-venv python3.11-dev python3-pip
```

Step 3: Create Swap File (t2.micro has only 1GB RAM)

```
sudo fallocate -l 2G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile
echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab

# Verify swap is active
free -h
```

Step 4: Create App User & Clone Repository

```
sudo useradd --system --create-home --shell /bin/bash quizapp
sudo git clone https://github.com/Nikhilramswaminaidu/youtube_video_quiz_generator.git /opt/youtube-quiz-gen
sudo chown -R quizapp:quizapp /opt/youtube-quiz-generator
```

Step 5: Set Up Python Virtual Environment

```
cd /opt/youtube-quiz-generator
sudo -u quizapp python3.11 -m venv venv
sudo -u quizapp /opt/youtube-quiz-generator/venv/bin/pip install --upgrade pip setuptools wheel
sudo -u quizapp /opt/youtube-quiz-generator/venv/bin/pip install -r requirements.txt
sudo -u quizapp /opt/youtube-quiz-generator/venv/bin/pip install gunicorn
```

Step 6: Configure Environment Variables

```
sudo nano /opt/youtube-quiz-generator/.env
```

Add the following content (replace the API key with your actual key):

```
# Required: Get your key at https://build.nvidia.com/  
NVIDIA_API_KEY=nvapi-your-actual-key-here  
  
# Optional: Cloudflare Worker URL for transcript proxying  
YOUTUBE_PROXY_URL=  
  
# Optional: Comma-separated custom Invidious instances  
INVIDIOUS_INSTANCES=
```

```
# Secure the .env file  
sudo chmod 600 /opt/youtube-quiz-generator/.env  
sudo chown quizapp:quizapp /opt/youtube-quiz-generator/.env
```

Step 7: Install Systemd Service

The systemd service runs the app automatically and restarts it on crashes:

```
sudo cp /opt/youtube-quiz-generator/deploy/aws/quiz-generator.service /etc/systemd/system/  
sudo systemctl daemon-reload  
sudo systemctl enable quiz-generator  
sudo systemctl start quiz-generator
```

Step 8: Install Nginx Configuration

This is the critical config that fixes the SSE connection timeout issue:

```
sudo cp /opt/youtube-quiz-generator/deploy/aws/nginx.conf /etc/nginx/sites-available/quiz-generator  
sudo rm -f /etc/nginx/sites-enabled/default  
sudo ln -sf /etc/nginx/sites-available/quiz-generator /etc/nginx/sites-enabled/  
sudo nginx -t && sudo systemctl reload nginx
```

7. Set Your API Key

Regardless of which deployment method you used, you MUST set your NVIDIA API key before the app will work.

Get Your NVIDIA NIM API Key

- Go to <https://build.nvidia.com/>
- Sign in or create a free account
- Search for 'Llama 3.3 70B Instruct'
- Click 'Get API Key' and copy it

```
A
# SSH into your instance
ssh -i ~/.ssh/quiz-generator-key.pem ubuntu@<YOUR_EC2_PUBLIC_IP>

# Edit the .env file
sudo nano /opt/youtube-quiz-generator/.env

# Replace the placeholder:
#   NVIDIA_API_KEY=nvapi-paste-your-key-here
# With your actual key:
#   NVIDIA_API_KEY=nvapi-xxxxxxxxxxxxxxxxxxxxxx

# Save (Ctrl+X, Y, Enter) and restart
sudo systemctl restart quiz-generator
```

Important

The app will NOT generate quizzes without a valid NVIDIA_API_KEY. The health endpoint (/api/health) will still respond, but quiz generation will fail with an authentication error.

8. Verify the Deployment

After setting your API key, verify everything is working:

```
C
sudo systemctl status quiz-generator
# Should show: active (running)
```

```
C
# From the server itself
curl http://localhost:8000/api/health

# From your local machine (replace IP)
curl http://<YOUR_EC2_PUBLIC_IP>/api/health
```

You should see a JSON response with yt-dlp version, python version, and status info.

```
C
http://<YOUR_EC2_PUBLIC_IP>/
```

You should see the YouTube Quiz Generator interface. Try generating a quiz!

Check Nginx is Proxying Correctly

```
sudo nginx -t          # Should show 'test is successful'
sudo systemctl status nginx # Should show 'active (running)'
```

9. HTTPS Setup (Optional)

The app works fine over HTTP. If you have a domain name, you can add free HTTPS with Let's Encrypt:

Step 1: Point Your Domain to the EC2 IP

In your DNS provider (Route 53, Cloudflare, Namecheap, etc.), create an A record pointing your domain to the EC2 instance's public IP.

```
# Example DNS record:
# Type: A
# Name: quiz (for quiz.yourdomain.com)
# Value: <YOUR_EC2_PUBLIC_IP>
```

Step 2: Update Nginx Server Name

```
# Edit the nginx config
sudo nano /etc/nginx/sites-available/quiz-generator

# Change: server_name _;
# To:      server_name quiz.yourdomain.com;

sudo nginx -t && sudo systemctl reload nginx
```

Step 3: Install SSL Certificate

```
# Install certbot if not already installed
sudo apt-get install -y certbot python3-certbot-nginx

# Get and install the certificate
sudo certbot --nginx -d quiz.yourdomain.com

# Certbot will automatically:
# - Generate an SSL certificate
# - Modify nginx config for HTTPS
# - Set up auto-renewal via systemd timer
```

Auto-Renewal

Certbot sets up automatic certificate renewal. You can verify it with:

```
sudo certbot renew --dry-run
```

10. Updating the App

To pull the latest code and restart the service:

```
# Option 1: Use the update script
sudo bash /opt/youtube-quiz-generator/deploy/aws/update.sh

# Option 2: Manual update
cd /opt/youtube-quiz-generator
sudo -u quizapp git pull origin main
sudo -u quizapp /opt/youtube-quiz-generator/venv/bin/pip install -r requirements.txt
sudo systemctl restart quiz-generator
```

11. Troubleshooting

A

```
# Check the service status
sudo systemctl status quiz-generator

# View detailed logs
sudo journalctl -u quiz-generator -n 100

# Follow live logs
sudo journalctl -u quiz-generator -f
```

502 Bad Gateway

This means nginx is running but can't reach the app. Check:

- Is the quiz-generator service running? `sudo systemctl status quiz-generator`
- Is gunicorn listening on port 8000? `curl http://localhost:8000/api/health`
- Check the app logs: `sudo journalctl -u quiz-generator -n 50`

Connection Refused / Timeout from Browser

This usually means the security group is blocking HTTP traffic:

- Go to AWS Console > EC2 > Security Groups
- Find the security group attached to your instance
- Add an inbound rule: Type=HTTP, Port=80, Source=0.0.0.0/0

NVIDIA API Errors (401 Unauthorized)

Your NVIDIA_API_KEY is missing or invalid.

```
# Check the .env file
cat /opt/youtube-quiz-generator/.env

# Make sure NVIDIA_API_KEY is set (not the placeholder)
# Restart after editing
sudo systemctl restart quiz-generator
```

y

```
# Reinstall yt-dlp in the virtual environment
sudo -u quizapp /opt/youtube-quiz-generator/venv/bin/pip install yt-dlp
sudo systemctl restart quiz-generator
```

Out of Memory

t2.micro only has 1GB RAM. Check if swap is active:

```
free -h
# Should show Swap: 2.0G

# If swap is missing, create it:
sudo fallocate -l 2G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile
```

12. Common Issues & Fixes

Problem	Likely Cause	Fix
502 Bad Gateway	App not running	<code>sudo systemctl start quiz-generator</code>
Connection timeout	Security group	Add HTTP (80) inbound rule
401 NVIDIA API error	Missing/wrong API key	Edit <code>.env</code> , restart service
yt-dlp not found	Not in venv PATH	<code>pip install yt-dlp</code> in venv
Out of memory	t2.micro = 1GB RAM	Check swap: <code>free -h</code>
App crashes on start	Python deps missing	<code>pip install -r requirements.txt</code>
SSE connection drops	nginx buffering	Check <code>proxy_buffering</code> off in config
Quiz generation slow	Normal for LLM	Expect 30-120s per batch

13. Useful Commands Reference

```
# == Service Management ==
sudo systemctl start quiz-generator      # Start the app
sudo systemctl stop quiz-generator       # Stop the app
sudo systemctl restart quiz-generator    # Restart the app
sudo systemctl status quiz-generator     # Check status

# == Logs ==
sudo journalctl -u quiz-generator -f    # Follow live logs
sudo journalctl -u quiz-generator -n 100 # Last 100 lines
sudo journalctl -u quiz-generator --since today

# == Nginx ==
sudo nginx -t                          # Test config
sudo systemctl reload nginx            # Reload config
sudo systemctl restart nginx           # Full restart

# == Health Check ==
curl http://localhost:8000/api/health  # From server
curl http://<EC2_IP>/api/health        # From outside

# == Update the App ==
sudo bash /opt/youtube-quiz-generator/deploy/aws/update.sh

# == Edit Environment Variables ==
sudo nano /opt/youtube-quiz-generator/.env
sudo systemctl restart quiz-generator

# == EC2 Instance IP ==
curl http://169.254.169.254/latest/meta-data/public-ipv4
```